

Analysis of Weather Simulation Algorithms Using Asymptotic Notation

Aim

To analyze and compare two weather simulation algorithms having time complexities $O(n(\log n)^2)$ and $O(n^2 \log n)$ using asymptotic notation and determine the most scalable algorithm for large-scale simulations.

Objective

- To study asymptotic notation.
- To compare the growth rates of $O(n(\log n)^2)$ and $O(n^2 \log n)$.
- To evaluate their performance for increasing input sizes.
- To determine the most efficient algorithm for weather simulations.

Theory

Weather simulation systems process a large number of atmospheric data points. As the dataset grows, the algorithm's time complexity becomes critical.

1. $O(n(\log n)^2)$

- Nearly linear growth with additional logarithmic factors.
- Suitable for processing large datasets efficiently.
- Performs well even as the number of weather data points increases.

2. $O(n^2 \log n)$

- Quadratic growth combined with a logarithmic factor.
- Processing time increases rapidly for large datasets.
- Less suitable for large-scale weather simulations.

Algorithm

1. Read the number of input sizes.
2. For each input:
 - Read n .
 - Compute $\log_2(n)$.

5. Print the conclusion.

Enter input size 3: 1000

1000 99316.86 9965784.28

Enter input size 4: 10000

10000 1765633.00 1328771237.95

Conclusion:

$O(n(\log n)^2)$ grows much slower than $O(n^2 \log n)$.

Therefore, $O(n(\log n)^2)$ is more scalable and efficient for large-scale weather simulations.

Analysis:

The program computes the theoretical values of the two time complexities for different input sizes.

Input (n)	$O(n(\log n)^2)$	$O(n^2 \log n)$
10	110.35	332.19
100	4414.08	66438.56
1000	99316.86	9965784.28
10000	1765615.34	1328771237.95

Observation

- $O(n(\log n)^2)$ grows slowly because only the logarithmic term is squared.
- $O(n^2 \log n)$ grows much faster due to the quadratic (n^2) factor.
- The performance gap increases significantly as the input size becomes larger.

For example:

- At $n = 100$, the second algorithm already requires about **15 times** more operations.
- At $n = 1000$, it requires about **100 times** more operations.
- At $n = 10000$, the difference becomes extremely large, making $O(n^2 \log n)$ unsuitable for large-scale simulations.

Growth Rate Comparison

Complexity	Growth Rate	Scalability
$O(n(\log n)^2)$	Moderate	Excellent
$O(n^2 \log n)$	Very High	Poor

Time Complexity

Algorithm	Time Complexity
Algorithm 1	$O(n(\log n)^2)$
Algorithm 2	$O(n^2 \log n)$

Space Complexity

Space Complexity = $O(1)$

Growth Rate Difference

The first algorithm increases almost linearly with additional logarithmic computations, while the second algorithm increases quadratically. As the input size becomes very large, the quadratic term dominates the execution time, causing $O(n^2 \log n)$ to become significantly slower.

Therefore, $O(n(\log n)^2)$ provides much better scalability and is more appropriate for weather simulation systems that process millions of data points.

Documentation

Software Requirements

- Turbo C / Dev C++ / Code::Blocks / VS Code
- GCC Compiler
- C Programming Language

Header Files Used

- `stdio.h` – Input and Output operations.
- `math.h` – Used for logarithmic calculations.

Functions Used

- `scanf()`
- `printf()`
- `log()`